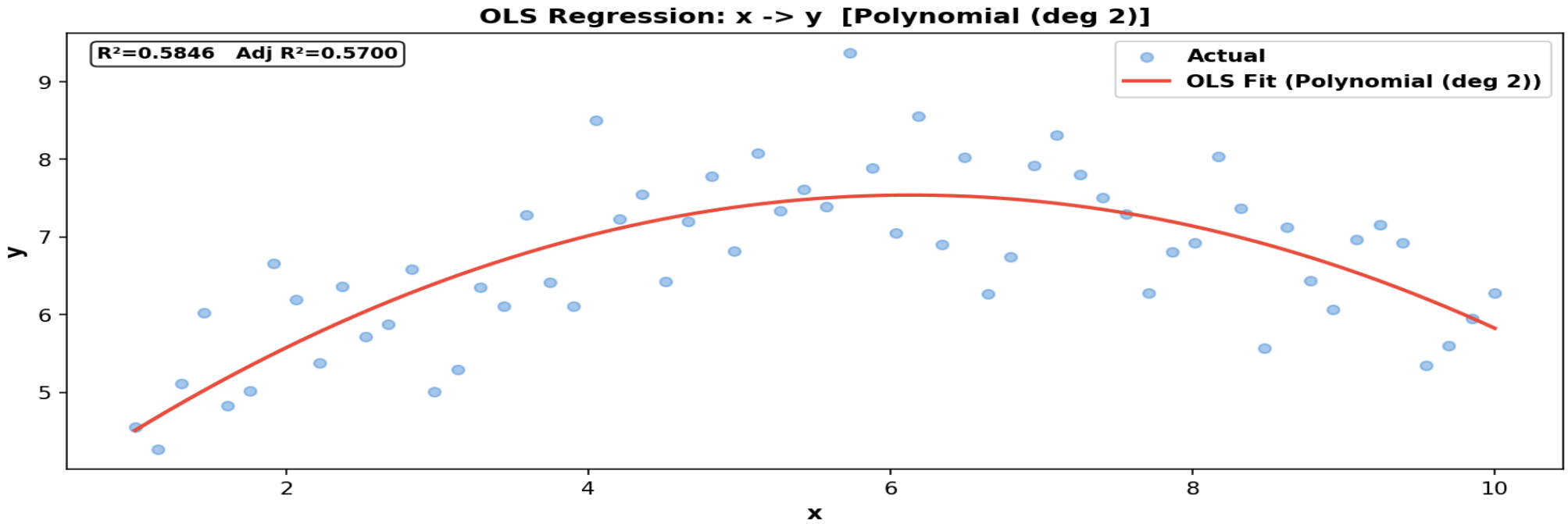
OLS Regression - Diagnostic Audit Report

Target: **y** | Predictors: **x** | Best Model: **Polynomial (deg 2)** | n = **60**

OLS Regression Plot - Best Model (Polynomial (deg 2))



Diagnostic Scorecard - Best Model (Polynomial (deg 2))

Test	Statistic	p-value	Threshold	Result	Notes
F-Test (Overall)	40.1004	0.0000	p < 0.05	PASS	df model=2, resid=57
Normality - Shapiro-Wilk	0.9876	0.8001	p > 0.05	PASS	Tests residual normality
Normality - Jarque-Bera	0.3269	0.8492	p > 0.05	PASS	Skew=0.162, Kurt=2.840
Homoscedasticity - Breusch-Pagan	1.4288	0.4895	p > 0.05	PASS	Tests constant variance
Autocorrelation - Durbin-Watson	2.0598	-	1.5 – 2.5	PASS	Tests residual independence
Multicollinearity - Max VIF	22.69	-	< 5	FAIL	Predictors: x, x^2
Influential Obs - Cook's D	2 flagged	-	4/n = 0.0667	FAIL	Red circles on residual plot

OLS Regression - Interpretation & Model Comparison

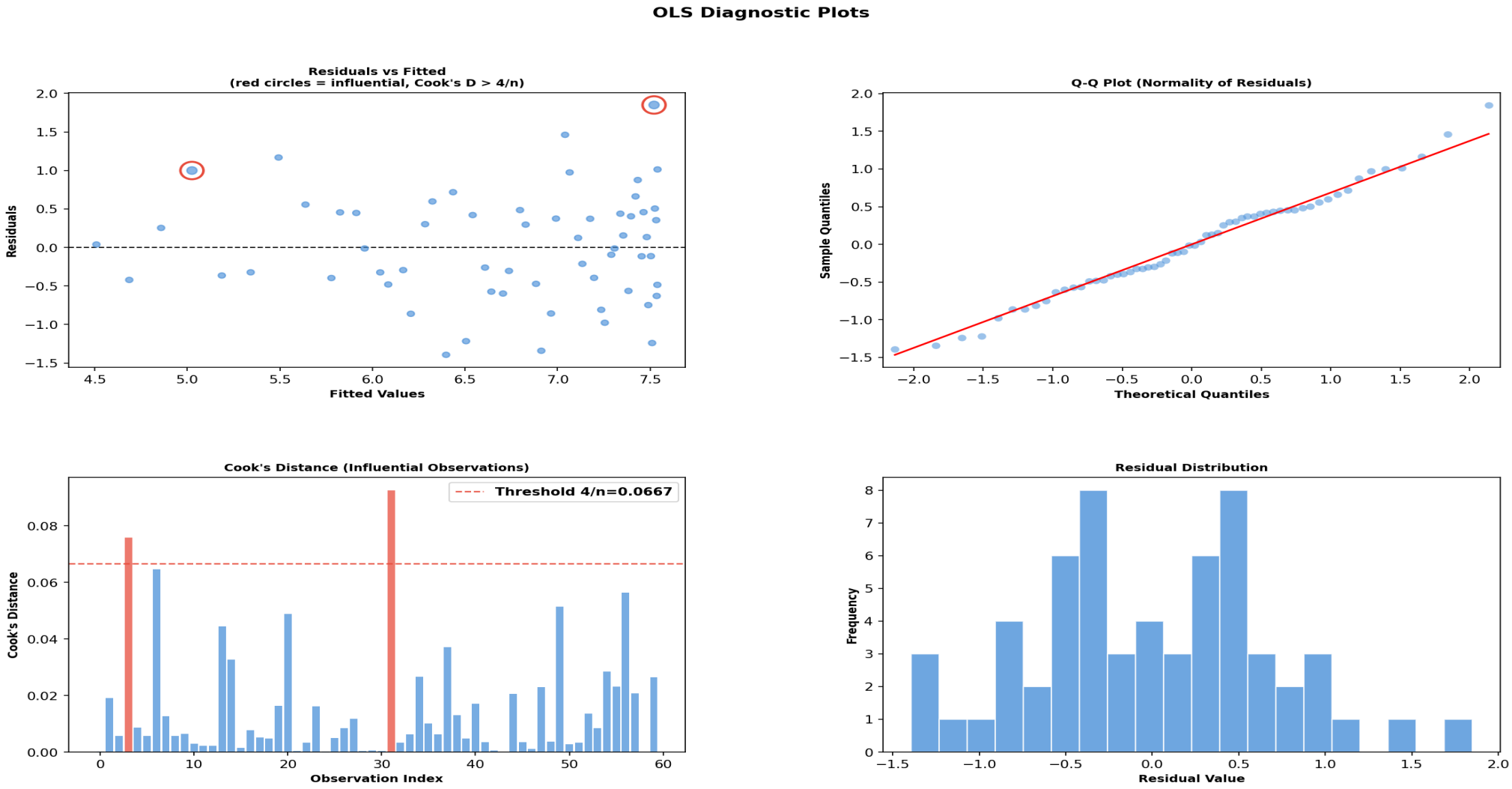
Target: **y** | Predictors: **x** | Best Model: **Polynomial (deg 2)** | n = **60**

Interpretation of Results

Overall Fit	The best model (degree 2) explains 58.5% of variance in y (R^2 = 0.5846, Adj R^2 = 0.5700), indicating a moderate fit.
Model Significance (F-Test)	F(2, 57) = 40.100, p = 0.0000. The overall regression is statistically significant ($p < 0.05$).
Normality of Residuals (Shapiro-Wilk)	W = 0.9876, p = 0.8001. Residuals appear normally distributed - the normality assumption is met.
Homoscedasticity (Breusch-Pagan)	BP stat = 1.4288, p = 0.4895. Residual variance appears constant (homoscedastic).
Autocorrelation (Durbin-Watson)	DW = 2.0598. No significant autocorrelation detected.
Multicollinearity (VIF)	Max VIF = 22.69 ≥ 5. Multicollinearity may inflate standard errors.
Influential Observations (Cook's Distance)	2 influential observation(s) detected (Cook's D > 0.0667). Review flagged points for data entry errors or genuine outliers.

Top 10 Models Ranked by BIC

Models ranked by BIC within each family (polynomial vs log/exponential) - lower BIC = better fit penalised for complexity. Cross-family comparison uses original-scale R^2 since BIC is not comparable across different response variable transformations. SW p = Shapiro-Wilk p-value (normality); BP p = Breusch-Pagan p-value (homoscedasticity). Green = PASS ($p > 0.05$), Red = FAIL ($p \leq 0.05$).

Rank	Model Type	Fitted Equation	R^2	Adj R^2	AIC	BIC	SW p	BP p
1	**Polynomial (deg 2)**	y = 3.214 + 1.4102*x - 0.1149*x^2	0.5846	0.57	131.15	137.43	**0.8001**	**0.4895**
2	**Polynomial (deg 3)**	y = 3.560 + 1.1320*x - 0.0562*x^2 - 0.0036*x^3	0.5869	0.5648	132.81	141.19	**0.8219**	**0.7214**
3	**Spline (Natural Cubic)**	y = 5.346 - 0.4516*cr(x, knots=[np.float64(3.25), np.float64(5.5), np.float64(7.750000000000001)])[0] + 1.0023*cr(x, knots=[np.float64(3.25), np.float64(5.5), np.float64(7.750000000000001)])[1] + 2.3591*cr(x, knots=[np.float64(3.25), np.float64(5.5), np.float64(7.750000000000001)])[2] + 1.8080*cr(x, knots=[np.float64(3.25), np.float64(5.5), np.float64(7.750000000000001)])[3] + 0.6277*cr(x, knots=[np.float64(3.25), np.float64(5.5), np.float64(7.750000000000001)])[4]	0.6024	0.5734	132.52	142.99	**0.7219**	**0.9513**
4	**Polynomial (deg 4)**	y = 4.736 - 0.2112*x + 0.4130*x^2 - 0.0667*x^3 + 0.0029*x^4	0.5949	0.5655	133.64	144.11	**0.8187**	**0.8747**
5	**Reciprocal**	y = 7.536 - 3.2329*1/x	0.3599	0.3489	155.09	159.28	**0.9671**	**0.1138**
6	**Logarithmic**	y = 5.280 + 0.9099*ln(x)	0.2665	0.2538	163.26	167.45	**0.9794**	**0.1487**
7	**Log-Log (Power Law)**	ln(y) = 1.655 + 0.1500*ln(x)	0.3033	0.2912	-63.9	-59.71	**0.8654**	**0.3536**
8	**Lag-1**	y = 5.949 + 0.5183*x - 0.3891*x_lag1	0.1048	0.0891	169.15	173.31	**0.8277**	**0.9701**
9	**Linear**	y = 5.888 + 0.1463*x	0.1316	0.1166	173.39	177.58	**0.811**	**0.7388**
10	**Exponential**	ln(y) = 1.754 + 0.0244*x	0.1531	0.1385	-52.18	-48.0	**0.6522**	**0.825**

Diagnostic Plots - Best Model